HSI-MFormer: Integrating Mamba and Transformer Experts for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is fundamental to numerous remote sensing applications, enabling detailed analysis of material properties and environmental conditions. Recent Mamba built upon selective state space models (SSMs) (S6) have demonstrated exceptional advantages in long-range sequence modeling with linear computational efficiency, while Transformer based on self-attention mechanisms is particularly adept at capturing short-range dependencies. To leverage the complementary strengths of these models, this article introduces a novel hybrid Mamba-Transformer framework (HSI-MFormer), effectively exploring the multiscale properties of hyperspectral data for HSI classification. Initially, a multiscale token generation (MTG) module is developed, which converts the HSI cube into multiple spatial–spectral token groups across different scales. To adequately capture fine-grained multiscale spatial–spectral patterns, an inner-scale transformer expert (ITE) is designed, which incorporates grouped self-attention operations to perform short-range sequence modeling within token groups at each scale. Meanwhile, a cross-scale Mamba expert (CME) is introduced, which integrates a cross-scale serialization mechanism and bidirectional Mamba block for long-range sequence modeling, further exploring the interactions and complementarity between token groups across different scales. Several hybrid strategies for integrating the ITE and CME are investigated to maximize their complementarity, including parallel, interval, and serial structures. Extensive experiments demonstrate that the proposed HSI-MFormer significantly outperforms the state-of-the-art Transformer-based and Mamba-based HSI classification methods. The code is available at https://github.com/tubingnuist/HSI-MFormer.**

***Index Terms*— Hyperspectral image (HSI) classification, Mamba, Transformer.**

Received 20 December 2024; revised 6 February 2025 and 5 April 2025; accepted 20 April 2025. Date of publication 24 April 2025; date of current version 5 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271200 and in part by the Start-Up Foundation for Introducing Talent of Nanjing University of Information Science and Technology (NUIST) under Grant 2023r091. *(Corresponding author: Bing Tu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3564167

I. Introduction

HYPERSPECTRAL image (HSI) classification has emerged as fundamental research in remote sensing fields, driven by the rich spectral and spatial information in HSI data. By analyzing hundreds of contiguous spectral bands across the electromagnetic spectrum, HSI classification enables precise material identification at the pixel level, capturing subtle variations in composition and structure. This technique underpins diverse remote sensing applications, ranging from environmental monitoring and precision agriculture [1], [2], [3] to urban planning [4], [5] and military surveillance [6], [7].

With the advancement of deep learning techniques, convolutional neural networks (CNNs) have garnered significant attention in HSI classification [8], [9], [10], [11], [12], [13]. Hu et al. [14] were the first to propose a hierarchical 1-D CNN architecture, effectively capturing high-level spectral features along the spectral dimension of hyperspectral data. Recognizing the strong spatial correlations inherent in HSIs, Lee and Kwon [15] introduced a contextual deep 2-D CNN architecture composed of multiple 2-D convolutional layers. While this approach effectively captures local spatial patterns within individual spectral bands, it fails to fully leverage the correlations between different spectral bands, potentially limiting its capacity to exploit the rich spectral information inherent in HSIs. To address this limitation, Hamida et al. [16] developed a 3-D CNN architecture capable of capturing more intricate spectral–spatial representations through multiple stacked 3-D convolutional kernels. By treating hyperspectral data as a volume rather than a collection of independent 2-D images, 3-D CNNs are better equipped to capture the complex interactions between spectral bands and spatial structures, particularly in scenarios with high variability in both spatial and spectral characteristics. In addition, Roy et al. [17] introduced a hybrid spectral architecture (HybridSN) that combines multiple spectral–spatial 3-D CNN blocks with spatial 2-D CNN blocks, effectively capturing fine-grained spatial–spectral patterns while reducing computational complexity. As a predominant backbone architecture, CNN-based models demonstrate remarkable capability in local feature extraction; however, they typically face challenges in capturing global dependencies between pixels.

Benefiting from the attention mechanism, Transformer architectures progressively demonstrate predominant status in

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HSI classification [18], [19], [20], [21], [22], [23]. Classically, He et al. [24] proposed a bidirectional encoder representation Transformer network (HSI-BERT). Based on the multihead self-attention (MHSA) mechanism, HSI-BERT is capable of learning multiple attention heads to encode context-aware semantic representations, thereby capturing discriminative spatialCspectral features. Hong et al. [25] reconsidered the HSI classification from a spectral sequential perspective and pioneeringly constructed a pure Transformer-based SpectralFormer (SF) network. Beyond bandwise representations in conventional Transformers, SF is dedicated to capturing the spectral sequence information between adjacent hyperspectral bands, thus yielding group-wise spectral embeddings. Considering the multiscale similarity and anisotropic image features of spectral data, Hu et al. [26] introduced a new multiscale multiangle attention network (MMAN) for HSI classification. This model combines a multiscale CNN network and a multiangle Transformer structure, effectively modeling the internal relationship between image features at local and global scales. To comprehensively integrate both spatial and spectral information for HSI classification, Peng et al. [27] introduced a dual-branch spatial–spectral Transformer with a cross-attention mechanism (CASST). The spatial branch is responsible for capturing fine-grained spatial information, and the spectral branch focuses on establishing interdependencies among spectral sequences. In addition, He et al. [23] investigated a hybrid multiscale spatial–spectral framework (HMSSF). It models global dependencies across multiple scales from both spatial and spectral domains, which achieves the cooperative extraction of hybrid spatial and spectral characteristics for HSI classification. Although Transformers have demonstrated substantial efficacy in capturing dependencies, the computational complexity of intrinsic attention mechanisms increases quadratically with sequence length. Therefore, Transformers are generally advantageous for modeling short-sequence variations [28], while exhibiting notable limitations to long-sequence ranges.

Recently, Mamba [29], building upon state space models (SSMs), has emerged as a promising alternative for sequence modeling. By incorporating selection mechanisms and hardware-aware implementation, Mamba achieves linear complexity scaling with sequence length, demonstrating competitive performance in long-range modeling compared to Transformers. Benefiting from its modeling capability and scalability, the Mamba architecture has been extensively adopted in remote sensing applications, including HSI classification [30], [31], [32], [33], [34], [35], [36], [37], [38]. For instance, Yang et al. [39] presented HSIMamba, a novel framework designed for efficient HSI classification. The model benefits from the state-space models (SSMs) with a bidirectional processing mechanism, which enables efficient extraction of spectral and spatial features from hyperspectral data while maintaining computational efficiency. Li et al. [30] proposed a novel image-level HSI classification model based on Mamba, named MambaHSI. To comprehensively leverage both spatial and spectral information, this model constructs a spatial and spectral mamba block to separately model the long-range interactions between spatial pixels and spectral bands. Meanwhile, a spatial–spectral fusion module is employed to adaptively integrate the information from both domains. Yao et al. [40] introduced SpectralMamba, an advanced HSI classification framework that addresses the challenges of spectral variability, redundancy, and computational efficiency from the perspective of sequence modeling. The framework integrates Piecewise sequential scanning (PSS) and gated spatial–spectral merging (GSSM) to encode the underlying spatial regularity and spectral peculiarity, yielding robust discriminative representations. Liu et al. [31] introduced a novel spectral–spatial adaptive Mamba for HSI classification (HyperMamba). This model integrates a spatial neighborhood adaptive scanning (SNAS) module and a spectral adaptive enhancement scanning (SAES) module, effectively encoding local spatial neighborhood information and spectral characteristics, which enhances spatial–spectral interactions while ensuring computational efficiency. Considering the high dimensionality of spectral data, He et al. [32] investigated a novel 3D-spectral–spatial mamba framework (3DSS-Mamba) for global spectral–spatial relationship modeling. A 3D-spectral–spatial selective scanning (3DSS) mechanism is introduced to perform pixel-wise selective scanning on 3-D hyperspectral tokens along the spectral and spatial dimensions, and five scanning routes are constructed to investigate the impact of dimension prioritization. In addition, Huang et al. [35] explored the potential of Mamba in the cross-scene classification of HSI. The proposed dynamic token augmentation mamba (DTAM) integrates a dynamic token augmentation module with a bidirectional scanning mechanism, effectively capturing the global features across different image patches.

Drawing inspiration from the benefits of hybrid architectures in language modeling [41], this work proposes a novel hybrid Mamba-Transformer architecture for HSI classification, denoted as HSI-MFormer. This framework consists of a multiscale token generation (MTG) module, an inner-scale Transformer expert (ITE), a cross-scale Mamba expert (CME), and a classifier. Specifically, the MTG employs multiscale 3-D convolutional operations to convert the HSI cube into multiple spatial–spectral token groups across different scales. These groups are subsequently reshaped into short sequences and long sequences in the inner-scale and cross-scale manner, respectively. The ITE incorporates grouped self-attention operations for short-range sequence modeling within token groups at each scale, effectively capturing multiscale spatial–spectral patterns while minimizing computational complexity. The CME built upon SSMs introduces a bidirectional Mamba block for long-range sequence modeling, further exploring the interactions and complementarity between token groups across different scales. Several hybrid strategies for integrating ITE and CME are investigated to maximize their complementarity, including parallel, interval, and serial structures. To the best of our knowledge, HSI-MFormer is the first effort to integrate the Mamba and Transformer for HSI classification. The main contributions are summarized as follows.

1) A hybrid Mamba-Transformer architecture HSI-MFormer is proposed for HSI classification, which achieves multiscale spectral–spatial feature extraction

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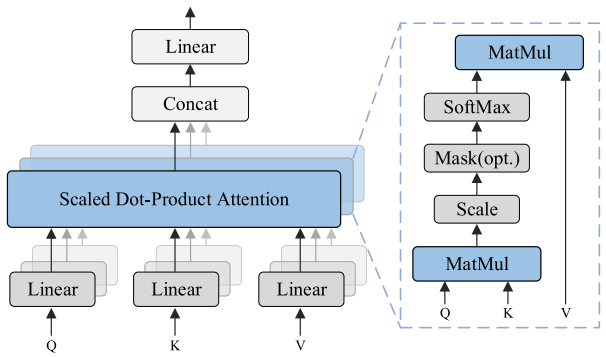


Fig. 1. Architecture of MHSA.

by leveraging the short-range and long-range modeling capabilities of Transformer and Mamba.

2) Several hybrid strategies for combining the ITE and CME experts are investigated. By identifying the optimal hybrid structure, the complementary strengths of Transformer and Mamba can be maximized.
3) Extensive experiments verified on four public hyperspectral datasets: Indian Pines, Pavia University, Houston 2013, and WHU-Hi-LongKou demonstrate the effectiveness and superiority of the proposed method.

The remaining sections of this article are organized as follows. Section II reviewed the preliminaries of Transformer and Mamba. Section III provides a comprehensive description of the proposed framework. Section IV outlines the experimental results and analyses. Conclusions and future work are discussed in Section V.

II. Preliminaries

A. Multihead Self-Attention

This structure of the multihead attention mechanism is illustrated in Fig. 1. By projecting the input tokens into multiple attention heads, the model can effectively capture complex dependencies within the data. Assuming the input feature matrix is denoted as $A \in \mathbb{R}^{H\times W\times C}$, where C represents the number of channels, and H and W correspond to the spatial dimensions, linear transformations are first applied to A to generate three distinct matrices: the query matrix Q, the key matrix K, and the value matrix V, as specified by the following:

$$Q = A \times W_Q, \quad K = A \times W_K, \quad V = A \times W_V \tag{1}$$

where W_Q, W_K, and W_V are learnable weight matrices of dimensions $C \times d_k$, with d_k being the dimension of the attention space. Next, the self-attention calculation process is performed on these transformed features Q, K, and V with the scaled dot-product attention, which can be formulated as

$$\mathrm{SA} = \mathrm{Attention}(Q, K, V) = \mathrm{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{2}$$

The MHSA mechanism involves mapping multiple sets of weight matrices W_Q, W_K, and W_V to calculate multihead attention values. The attention values for each head are computed independently, and the outputs of all heads are then concatenated together along the feature dimension. The calculation process is as follows:

$$\mathrm{MHSA}(Q, K, V) = \mathrm{Concat}(\mathrm{SA}_1, \ldots, \mathrm{SA}_i, \ldots, \mathrm{SA}_h)W \tag{3}$$

where h is the number of heads, and W represents the learned parameter matrices.

Arising from the necessity of computing pairwise interactions between all elements in the sequence, the self-attention mechanism exhibits quadratic computational complexity with respect to the sequence length. The computational complexity can be described as

$$\mathrm{Complex(MHSA)} = 4HWC^2 + 2(HW)^2C. \tag{4}$$

With the attention mechanism as a core component, the Transformer inevitably suffers from substantial computational and memory demands when processing long sequences, presenting challenges in applications requiring efficient modeling of extended temporal or spatial dependencies.

B. State Space Models

Building upon the principles of linear time-invariant systems, SSMs are formulated to map a one-dimensional input signal $x(t) \in \mathbb{R}$ into an output sequence $y(t) \in \mathbb{R}$ via an intermediate hidden state $h(t) \in \mathbb{R}^N$. This transformation can be mathematically described through the following linear ordinary differential equation (ODE):

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{5}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition parameter, and $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{N\times 1}$ is the projection matrices.

To integrate the continuous-time system depicted in (5) into discrete sequence-based deep models, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ are subsequently discretized via a zero-order hold (ZOH) technique with a time scale parameter Δ. This process can be mathematically represented as

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \end{aligned} \tag{6}$$

where $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ denote the discretized forms of the parameters $\mathbf{A}$ and $\mathbf{B}$, respectively. Subsequently, the discretized SSM system can be expressed as

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t + \mathbf{D}x_t. \end{aligned} \tag{7}$$

For efficient implementation, the aforementioned linear recurrence process is achieved through the following convolution operation, which can be expressed as:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{\mathbf{L}-1}\overline{\mathbf{B}}\right) \\ \mathbf{y} &= \mathbf{x} * \overline{\mathbf{K}} \end{aligned} \tag{8}$$

where L denotes the length of input sequence, and $\overline{\mathbf{K}} \in \mathbb{R}^L$ serves as the structured convolutional kernel.

Mamba [29] introduces the selective SSMs (S6), which enables the projection matrices to be modified as

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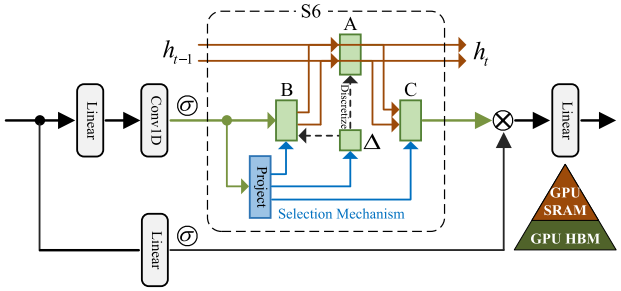


Fig. 2. Architecture of the Mamba block.

input-dependent to achieve selective attention on each sequence unit. This structure is illustrated in Fig. 2. Specifically, it involves defining the interval $\mathbf{\Delta}$, and matrices B, C as functions of the input $x \in \mathbb{R}^{B\times L\times D}$, which can be formulated as

$$\begin{aligned}\mathbf{B} &= \text{Projection}_B(x)\\ \mathbf{C} &= \text{Projection}_C(x)\\ \mathbf{\Delta} &= \tau_\Delta\big(\text{Parameter} + \text{Projection}_\Delta(x)\big)\end{aligned} \tag{9}$$

where $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$ and $\mathbf{\Delta} \in \mathbb{R}^{B\times L\times D}$. The functions $\text{Projection}_B(\cdot)$ and $\text{Projection}_C(\cdot)$ perform linear projections to the N-dimensional space, and the symbol τ_Δ represents the Softplus activation function.

Building on the aforementioned state transition procedure, Mamba exhibits linear computational complexity and selective information processing, conferring notable advantages in long-range sequence dependency modeling. As referenced in [42], the computational complexity can be expressed as

$$\text{Complex(Mamba)} = 3L(2D)N + L(2D)N. \tag{10}$$

Mamba is linear to the sequence length L. The state dimension N is a fixed parameter, set to 16 by default.

III. Proposed Network

The architecture of the proposed HSI-MFormer for HSI classification is depicted in Fig. 3, which consists of an MTG, an ITE, a CME, and a Classifier. The MTG processes the cropped input patch, converting it into multiple parallel spatial–spectral token groups across varying scales. The ITE expert, which operates based on the grouped self-attention mechanism, and the CME expert, which employs bidirectional SSMs, capture multiscale spatial–spectral information from short-range and long-range modeling perspectives, respectively. Finally, the Classifier integrates the complementary information extracted by the preceding components to perform classification.

A. Multiscale Token Generation Module

Traditional HSI classification methods typically perform data tokenization within a single scale, which inevitably restricts the models' capability to capture multiscale spatial–spectral details and structural information. To overcome this limitation, the proposed MTG module incorporates multiple 3-D convolutions with different kernel scales for MTG, which are subsequently transmitted to the Transformer and Mamba experts for short-range and long-range spatial–spectral feature extraction.

As shown in Fig. 3, MTG is constructed with three parallel 3-D convolution groups of small, medium, and large scales. Each group consists of a 3-D convolution layer (3DConv), a batch normalization (BN) layer, and a rectified linear activation function (ReLU). Taking the cropped HSI patch x as input, the process can be expressed as

$$\begin{aligned}f_S &= \text{ReLU}(\text{BN}(\text{3DConv}_{s\times s\times s}(x)))\\ f_M &= \text{ReLU}(\text{BN}(\text{3DConv}_{m\times m\times m}(x)))\\ f_L &= \text{ReLU}(\text{BN}(\text{3DConv}_{l\times l\times l}(x)))\end{aligned} \tag{11}$$

where $s \times s \times s$, $m \times m \times m$, and $l \times l \times l$ represent the 3-D kernel sizes corresponding to the small, medium, and large convolution groups, respectively. The number of convolutional filters in all groups is set to 32.

Following each convolution group, MTG further involves a linear embedding layer for dimension transformation. This procedure can be formulated as

$$\begin{aligned}T_S &= \text{Linear}(f_S)\\ T_M &= \text{Linear}(f_M)\\ T_L &= \text{Linear}(f_L)\end{aligned} \tag{12}$$

where T_S, T_M, and T_L denote the generated spectral–spatial token groups with different scales of $S \times S \times C$, $M \times M \times C$, and $L \times L \times C$, respectively. Notably, the embedding operation is carried out independently for each group, without parameter sharing.

B. Inner-Scale Transformer Expert

The Transformer leverages the self-attention mechanism to access tokens and establish dependencies. While its quadratic computational complexity imposes significant challenges for efficiently handling long-sequence tasks, it remains advantageous for short-range dependency modeling. In light of this, the proposed ITE incorporates the grouped self-attention strategy to perform short-range dependency modeling within token groups at each scale, effectively capturing fine-grained multiscale spatial–spectral patterns while minimizing computational complexity.

As shown in Fig. 3, instead of integrating all token groups as a whole for processing, ITE treats each token group with different scales as an independent short sequence, which executes standard self-attention computations independently within each token group. To be specific, the generated token groups $T_S \in \mathbb{R}^{S\times S\times C}$, $T_M \in \mathbb{R}^{M\times M\times C}$, and $T_L \in \mathbb{R}^{L\times L\times C}$ are parallelly input into the grouped MHSA block for short-range dependency modeling, combined with a residual operation. The computation procedure can be formulated as

$$\begin{aligned}F_S &= T_S + \text{MHSA}_S(\text{LN}(T_S))\\ F_M &= T_M + \text{MHSA}_M(\text{LN}(T_M))\\ F_L &= T_L + \text{MHSA}_S(\text{LN}(T_S))\end{aligned} \tag{13}$$

where $\text{LN}(\cdot)$ refers to the layer normalization operation. The $\text{MHSA}_S(\cdot)$, $\text{MHSA}_M(\cdot)$, and $\text{MHSA}_L(\cdot)$ denote the standard

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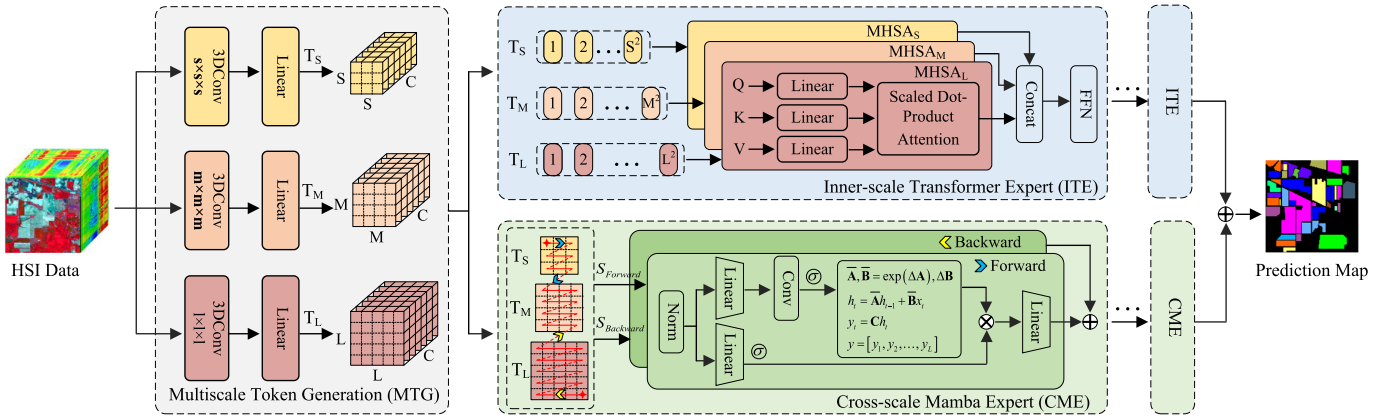


Fig. 3. Overall architecture of the hybrid Mamba-Transformer framework (HSI-MFormer), taking the parallel structure as an example.

MHSA mechanism customized for the corresponding token group, with the detailed computation referenced in (3).

After that, the output features of all groups are concatenated in sequence to comprehensively leverage the obtained information, as denoted by

$$F_{\text{ITE}} = \text{Concat}(F_S, F_M, F_L) \tag{14}$$

where Concat$(\cdot)$ represents the concatenation operation. Then, an FFN module is primarily utilized to perform nonlinear transformations, enhancing the model's representational capacity and performance

$$F_{\text{ITE}} = F_{\text{ITE}} + \text{FFN}(F_{\text{ITE}}) \tag{15}$$

where FFN$(\cdot)$ denotes the feed-forward network comprising two linear layers and an activation function.

Following (4), when applying the Transformer to integrate all the token groups as a whole for attention computation, the complexity can be expressed as:

$$\text{Complex}(T) = 4(S+M+L)^2C^2 + 2(S+M+L)^4C \tag{16}$$

which inevitably suffers from significant computation burdens.

In comparison, the computational complexity for ITE is described as the sum of grouped attention blocks, which can be formulated as

$$\begin{aligned}\text{Complex(ITE)} = {} & 4\left(S^2+M^2+L^2\right)C^2 \\ & + 2\left(S^4+M^4+L^4\right)C.\end{aligned} \tag{17}$$

Evidently, the proposed ITE significantly reduces computational complexity while effectively enabling multiscale spectral–spatial information extraction.

C. Cross-Scale Mamba Expert

Recent Mamba incorporates a selective mechanism to retain relevant information and filter out irrelevant information indefinitely. This selective mechanism with linear complexity property allows Mamba to preserve essential long-term patterns while disregarding transient fluctuations [43]. Building on this inspiration, this article introduces the CME, which performs long-range sequence modeling to further explore the interactions and complementarity between token groups across different scales.

As shown in Fig. 3, CME first integrates and flattens all token groups into a whole long-range sequence in the order of resolution scale. Both forward and backward scanning directions are considered to enhance the capacity for spatially-aware processing. Taking the generated token groups $T_S \in \mathbb{R}^{S\times S\times C}$, $T_M \in \mathbb{R}^{M\times M\times C}$, and $T_L \in \mathbb{R}^{L\times L\times C}$ as input, the flattening process can be formulated as

$$\begin{aligned} S_{\text{Forward}} &= \left[T_S^1, \ldots, T_S^{S^2}, T_M^1, \ldots, T_M^{M^2}, T_L^1, \ldots, T_L^{L^2}\right] \\ S_{\text{Backward}} &= \left[T_L^{L^2}, \ldots, T_L^1, T_M^{M^2}, \ldots, T_M^1, T_S^{S^2}, \ldots, T_S^1\right] \end{aligned} \tag{18}$$

where $S_{\text{Forward}} \in \mathbb{R}^{C\times(S^2+M^2+L^2)}$ indicates the generated forward sequence, and $S_{\text{Backward}} \in \mathbb{R}^{C\times(S^2+M^2+L^2)}$ represents the reversed sequence.

Subsequently, the flattened sequences are transmitted into a bidirectional Mamba block for independent long-range sequence modeling. For each direction, the sequence is first normalized and then passed through two parallel linear embedding layers: one branch is followed by an activation function for gating signal generation, while the other undergoes a 1-D convolution operation. Assuming $S \in \{S_{\text{Forward}}, S_{\text{Backward}}\}$, the procedure for each direction can be formulated as

$$z = \sigma(\text{Linear}(\text{Norm}(S))) \tag{19}$$

$$f = \sigma(\text{Conv}(\text{Linear}(\text{Norm}(S)))) \tag{20}$$

where σ denotes the Silu activation [44] operation. After this, the generated representation f is passed into the S6 model for state space evolution, followed by layer normalization and a gating operation. Finally, the features are fed to the ultimate linear layer with a residual connection

$$F = \text{Linear}(\text{Norm}(S6(f)) \otimes z) + S \tag{21}$$

where $\otimes$ denotes the matrix multiplication operation. The $S6(\cdot)$ model serves as the core unit, with its specific computation referenced in (7).

After parallel sequential processing, the generated features F_{Forward} and F_{Backward} are subsequently aggregated through vector addition, obtaining the final output as

$$F_{\text{CME}} = F_{\text{Forward}} + F_{\text{backward}}. \tag{22}$$

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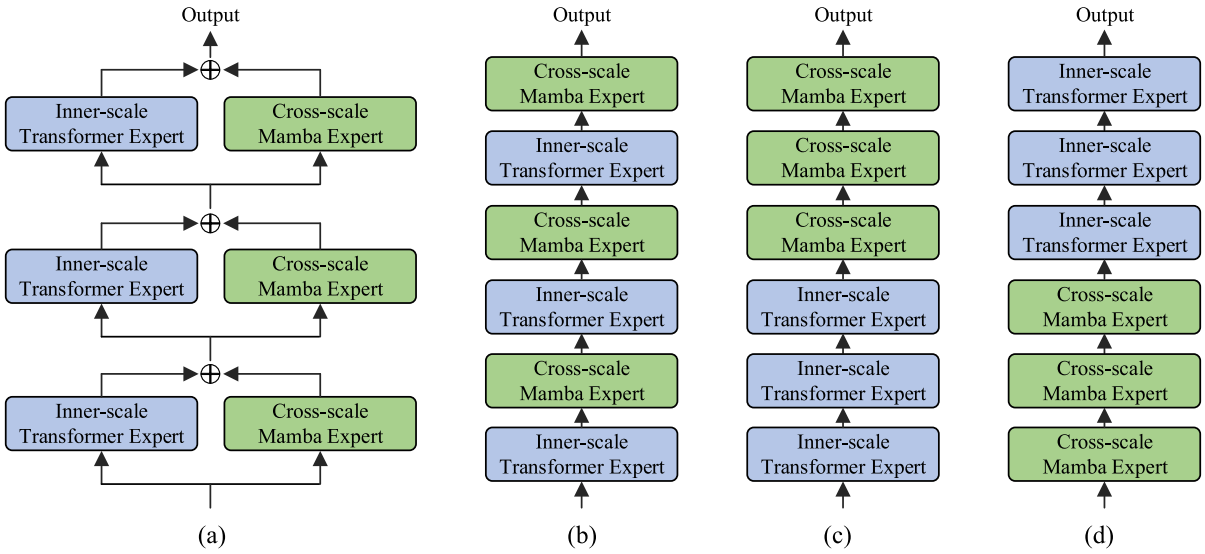


Fig. 4. Hybrid structures for HSI-MFormer. (a) Parallel structure. (b) Interval structure. (c) Serial structure v1. (d) Serial structure v2.

Benefiting from the cross-scale serialization strategy and long-range modeling property, the proposed CME provides an enhanced receptive field for hyperspectral data, facilitating the extraction of multiscale spatial–spectral patterns with linear computational complexity.

D. Hybrid Structure for HSI-MFormer

To fully leverage the complementary advantages of Transformer and Mamba, this article investigates a series of hybrid structures for combining the ITE and CME layers, as shown in Fig. 4.

1) Parallel Structure: The ITE and CME operate in parallel at each stage, with their outputs fused through direct addition. The depth for both ITE and CME is set to N. Consequently, this structure can be delineated as follows:

$$F_{i+1} = \mathrm{ITE}(F_i) + \mathrm{CME}(F_i) \quad \forall i = 1, 2, \ldots, N. \tag{23}$$

2) Interval Structure: In this configuration, the ITE and CME are sequentially stacked in a layer-wise interleaved manner, which can be expressed as

$$F_{i+1} = \mathrm{CME}(\mathrm{ITE}(F_i)) \quad \forall i = 1, 2, \ldots, N. \tag{24}$$

3) Serial Structure V1: In this structure, the ITE is deployed in the initial N layers, followed by the CME in the subsequent N layers. The process can be described as

$$F_{i+1} = \begin{cases} \mathrm{ITE}(F_i), & \text{if } i \leq N \\ \mathrm{CME}(F_i), & \text{if } N < i \leq 2N. \end{cases} \tag{25}$$

4) Serial Structure V2: In this configuration, the order of the CME and ITE experts is reversed, with N consecutive CME layers preceding N consecutive ITE layers

$$F_{i+1} = \begin{cases} \mathrm{CME}(F_i), & \text{if } i \leq N \\ \mathrm{ITE}(F_i), & \text{if } N < i \leq 2N. \end{cases} \tag{26}$$

The proposed HSI-MFormer allows for the flexibility to adopt the aforementioned structures, with the final output sent to the classifier for ultimate classification

$$\mathrm{pred} = \mathrm{Classifier}(F_{\mathrm{out}}) \tag{27}$$

where Classifier$(\cdot)$ consists of one perceptron layer with an embedding dimension of 32. The classification is optimized with the cross-entropy loss function. The performance of different architectures is comprehensively evaluated in the experimental section.

IV. Performance Evaluation

A. Datasets Description

To demonstrate the effectiveness of the proposed approach, extensive experimental comparisons are conducted on four publicly HSI databases: Indian Pines, Pavia University, Houston 2013, and WHU-Hi-LongKou [45]. Table I provided the division details regarding the training and testing sets.

1) Indian Pines: This dataset was collected by the Airborne/Visible Infrared Imaging Spectrometer (AVIRIS) imaging an Indian pine tree over Northwestern Indiana in 1992. The spectrometer's wavelength spans from 0.4 to 2.5 μm. After removing the water absorption channels, it includes 200 spectral bands and a spatial resolution of 20 m per pixel, covering an area of 145 $\times$ 145 pixels. There are 10 249 ground sample points, representing 16 distinct classes.

2) Pavia University: This dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over the city of Pavia in Northern Italy. The spectrometer's imaging wavelength ranges from 0.43 to 0.86 μm. After removing 12 noisy bands, it consists of 103 spectral bands and a spatial resolution of 1.3 m per pixel, covering an area of 610 $\times$ 340 pixels. There are a total of 42 776 labeled ground object pixels with 9 distinct categories.

3) Houston 2013: This dataset was collected by the ITRES CASI-1500 sensor over the University of Houston campus and surrounding areas, provided by the 2013 GRSS Data Fusion

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TABLE I

CATEGORY NAME, SAMPLE NUMBERS OF TRAIN SET AND TEST SET OF EACH CLASS ON THE INDIAN PINES, PAVIA UNIVERSITY, HOUSTON 2013, AND WHU-HI-LONGKOU DATASETS

	Indian Pines			**Pavia University**			**Houston 2013**			**WHU-Hi-LongKou**		
No.	Category	Train	Test	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Alfalfa	5	41	Asphalt	332	6299	Healthy grass	125	1126	Corn	345	34166
2	Corn-notill	143	1285	Meadows	932	17717	Stressed grass	125	1129	Cotton	84	8290
3	Corn-mintill	83	747	Gravel	105	1994	Synthetic grass	70	627	Sesame	30	3001
4	Corn	24	213	Trees	153	2911	Trees	124	1120	Broad-leaf soybean	632	62580
5	Grass-pasture	48	435	Painted metal sheets	67	1278	Soil	124	1118	Narrow-leaf soybean	42	4109
6	Grass-trees	73	657	Bare soil	251	4778	Water	33	292	Rice	119	11735
7	Grass-pasture-mowed	3	25	Bitumen	67	1263	Residential	127	1141	Water	671	66385
8	Hay-windrowed	48	430	Self-blocking bricks	184	3498	Commercial	124	1120	Roads and houses	71	7053
9	Oats	2	18	Shadows	47	900	Road	125	1127	Mixed weed	52	5177
10	Soybean-notill	97	875				Highway	123	1104			
11	Soybean-mintill	245	2210				Railway	123	1112			
12	Soybean-clean	59	534				Parking Lot 1	123	1110			
13	Wheat	20	185				Parking Lot 2	47	422			
14	Woods	126	1139				Tennis Court	43	385			
15	Buildings	39	347				Running Track	66	594			
16	Stone	9	84									
	Total	1024	9225	Total	2138	40638	Total	1502	13527	Total	2046	202496

TABLE II

EFFECTIVENESS ANALYSIS OF DIFFERENT HYBRID STRUCTURES ON ALL DATASETS IN TERMS OF OA (%), AA (%), AND KAPPA (%)

	Indian Pines			**Pavia University**			**Houston2013**			**WHU-Hi-LongKou**		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Parallel Structure	99.01	98.91	98.87	99.67	99.38	99.57	99.31	99.37	99.25	99.59	98.58	99.46
Interval Structure	98.85	98.63	98.68	99.57	99.21	99.43	99.26	99.36	99.20	99.54	98.41	99.41
Serial Structure v1	98.90	98.45	98.74	99.65	99.3	99.53	99.32	99.41	99.27	99.59	98.53	99.46
Serial Structure v2	98.87	98.30	98.71	99.4	98.86	99.21	99.26	99.41	99.27	99.51	98.34	99.35

Contest. The imaging wavelength of the spectrometer ranges from 0.38 to 1.05 μm. The dataset comprises 144 spectral bands and covers an area of 340 $\times$ 1905 pixels, with a spatial resolution of 2.5 m per pixel. It includes a total of 15 029 sample pixels, categorized into 15 challenging types.

4) WHU-Hi-LongKou: This dataset was acquired in Longkou Town, Hubei province, China, 2018, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. The imaging wavelength of the spectrometer ranges from 400 to 1000 nm. It comprises 270 spectral bands and covers an area of 550 $\times$ 400 pixels, with a spatial resolution of 0.463 m per pixel. This dataset is categorized into 9 challenging species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, rice, water, roads and houses, and mixed weed.

B. Experimental Settings

1) Evaluation Metrics: Following existing HSI classification protocols, the evaluation metrics including overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are leveraged for evaluation. All experiments are conducted under identical experimental conditions, with the reported results averaged over five consecutive trials.

2) Implementation Details: All experiments are implemented on the PyTorch platform with a single RTX 3070Ti GPU. The Adam optimizer is leveraged for parameter optimization, with a learning rate of 0.001. The training process runs for 100 iterations and the model at the 100th iteration is selected as the optimal model. The batch size is set to 64. The size of the cropped HSI patch is set to 15 $\times$ 15. The PCA dimension for reduction is set to 30. The kernel scales s, m, and l in MTG are set to 9, 7, and 5, respectively. The state dimension and expansion ratio in the Mamba branch are fixed at 16 and 1, respectively.

3) Competitive Approaches: Four kinds of representative HSI classification architectures are selected for comprehensive comparison, including conventional methods (SVM [46]), CNN-based methods (2D-CNN and 3D-CNN [16]), Transformer-based methods (HSI-BERT [24], SF [25], and CASST [27]), and Mamba-based methods (3DSS-Mamba [32], MambaHSI [30], and HyperMamba [31]).

C. Analysis to Different Hybrid Structures

This section explores the effectiveness of the constructed four hybrid structures, including Parallel Structure, Interval Structure, Serial Structure v1, and Serial Structure v2. The classification results on three datasets are illustrated in Table II. The exceptional classification performance across all structures highlights the superiority of HSI-MFormer in effectively integrating the strengths of Mamba and Transformer.

For the Indian Pines dataset, all hybrid structures achieve high OAs, with Serial Structure v1 achieving the highest value of 98.90%, followed closely by the Parallel Structure. In terms of AA, the Parallel Structure leads with 98.66%, while Kappa coefficients are comparable across methods, with Serial Structure v1 performing slightly better.

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TABLE III

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods			HSI-MFormer
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	CASST	3DSS-Mamba	MambaHSI	HyperMamba	
1	20.73	76.09	95.65	48.78	**100.0**	78.26	78.05	90.24	**100.0**	95.12
2	73.07	92.16	88.31	78.44	78.37	97.34	90.27	96.61	**97.66**	96.26
3	62.99	87.23	84.94	80.58	89.29	97.71	**93.04**	98.91	97.85	**99.73**
4	50.70	64.98	87.76	51.17	81.22	99.58	93.43	98.58	99.53	**100.0**
5	92.64	96.48	96.48	91.24	89.89	92.96	97.70	93.73	**100.0**	99.54
6	94.90	98.77	96.16	96.19	97.41	97.81	97.26	99.07	99.23	**99.85**
7	76.00	32.14	78.57	20.00	84.00	96.43	96.00	87.50	**100.0**	**100.0**
8	96.63	**100.0**	**100.0**	98.83	**100.0**	99.79	99.30	**100.0**	**100.0**	**100.0**
9	33.33	30.00	70.00	0.00	27.78	65.00	38.89	93.75	83.33	**100.0**
10	68.69	91.87	92.80	78.37	94.97	92.80	96.80	98.96	**98.97**	98.29
11	85.16	95.93	91.20	92.85	93.44	98.37	98.91	**99.72**	99.23	99.68
12	64.89	93.42	88.03	62.47	81.65	91.40	91.39	97.54	96.06	**98.13**
13	97.03	97.56	94.15	96.19	87.03	91.71	95.68	**100.0**	98.37	99.46
14	96.49	98.81	99.05	94.46	94.82	99.76	99.65	98.75	99.91	**100.0**
15	55.33	81.35	89.38	87.89	96.83	95.34	94.52	**100.0**	99.42	**100.0**
16	93.93	97.85	**100.0**	65.06	95.24	81.72	84.52	96.38	98.80	96.43
OA (%)	79.82	93.34	92.17	85.45	90.67	96.65	95.82	98.46	98.76	**99.01**
AA (%)	72.03	83.41	90.78	71.41	87.00	92.25	90.33	96.86	97.96	**98.91**
Kappa	76.84	92.39	91.08	83.36	89.36	96.18	95.23	97.66	98.59	**98.87**

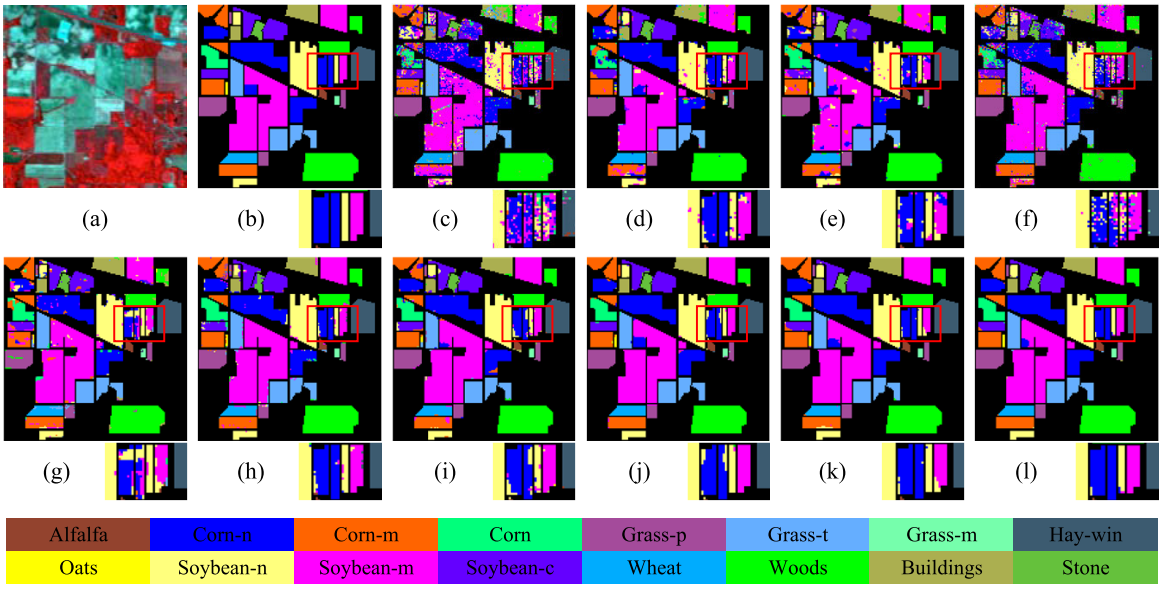


Fig. 5. Classification maps obtained by the compared methods on the Indian Pines dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) CASST. (i) 3DSS-Mamba. (j) MambaHSI. (k) HyperMamba. (l) HSI-MFormer.

On the Pavia University dataset, the Serial Structure v1 demonstrates the best performance in terms of OA (99.65%), AA (99.30%), and Kappa (99.53%). The Parallel Structure follows closely, with an OA of 99.64%, an AA of 99.28%, and a Kappa of 99.52%, indicating its robustness on this dataset.

For the Houston 2013 dataset, the Serial Structure v1 achieves the best OA of 99.32% and Kappa of 99.27%, while both Serial Structure v1 and v2 achieve the highest AA. Other hybrid structures exhibit minor variations but maintain high overall performance across all metrics.

On the WHU-Hi-LongKou dataset, the Parallel Structure demonstrates the best performance in terms of OA (99.59%), AA (99.58%), and Kappa (99.46%). The Serial Structure v1 follows closely, which yields a slightly lower AA of 99.53%. These results demonstrate the strong performance and robustness of the hybrid integration strategies, with Serial Structure v1 and Parallel Structure consistently achieving competitive results across all datasets.

D. Experimental Comparison With Competitive Approaches

The quantitative accuracies [OA (%), AA (%), and Kappa (%)] for the Indian Pines, Pavia University, Houston 2013, and WHU-Hi-LongKou datasets are summarized in Tables III–VI, with the optimal results highlighted in bold. The corresponding classification maps are shown in Figs. 5–8.

1) Indian Pines Dataset: The experiments on the Indian Pines dataset are performed with 10% of the reference samples. The corresponding quantitative classification accuracies and visualization maps are presented in Table III and Fig. 5.

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TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods			HSI-MFormer
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	CASST	3DSS-Mamba	MambaHSI	HyperMamba	
1	91.63	96.31	95.94	97.63	94.75	99.17	99.33	**99.87**	99.55	99.32
2	97.56	99.37	99.71	99.93	98.14	99.95	99.34	99.97	**100.0**	**100.0**
3	73.90	82.22	89.76	86.51	84.70	98.28	95.09	**99.59**	98.19	98.90
4	92.10	94.35	97.36	97.53	96.81	93.08	95.91	96.87	98.45	**99.11**
5	98.51	99.85	**100.0**	99.61	99.22	98.66	99.69	**100.0**	**100.0**	99.77
6	86.04	93.65	97.41	93.05	94.37	99.60	99.43	**100.0**	**100.0**	**100.0**
7	84.09	89.17	96.69	95.80	84.56	99.62	95.88	99.84	**100.0**	**100.0**
8	90.25	88.73	93.92	97.86	92.65	94.73	96.20	98.81	98.62	**99.43**
9	99.22	97.57	**100.0**	99.67	**100.0**	96.41	97.22	98.31	98.44	97.89
OA (%)	92.75	95.77	97.62	97.61	96.60	98.63	98.48	99.58	99.56	**99.67**
AA (%)	90.37	93.47	96.75	96.40	93.91	97.72	97.56	99.25	99.25	**99.38**
Kappa	90.35	94.37	96.84	96.82	94.17	98.19	97.98	99.37	99.44	**99.57**

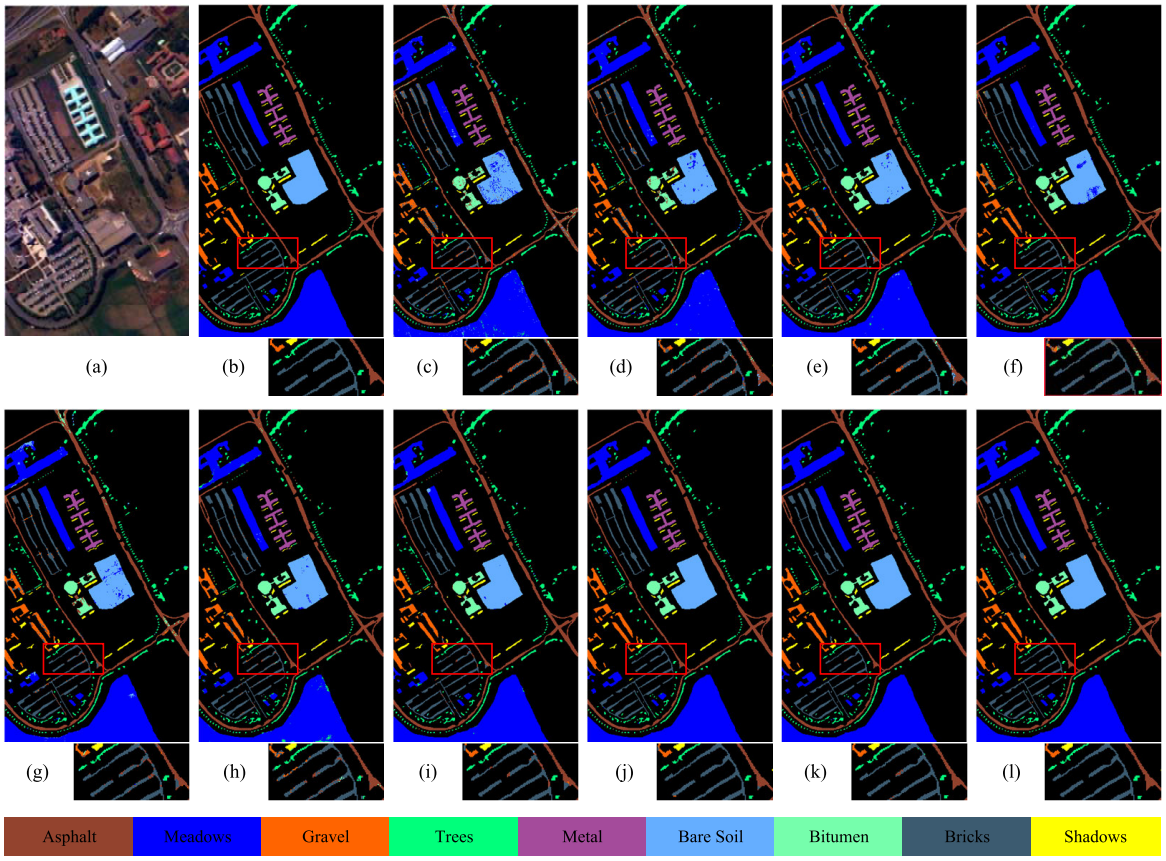


Fig. 6. Classification maps obtained by the compared methods on the Pavia University dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) CASST. (i) 3DSS-Mamba. (j) MambaHSI. (k) HyperMamba. (l) HSI-MFormer.

As observed from Table III, the proposed HSI-MFormer exhibits clear superiority and effectiveness over existing conventional, CNN-based, Transformer-based, and Mamba-based methods. Compared to the suboptimal HyperMamba method, HSI-MFormer demonstrates notable improvements in OA, AA, and Kappa by 0.25%, 0.95%, and 0.28%, respectively. Conventional and CNN-based methods, such as SVM and 2D-CNN, are inherently constrained by manually crafted descriptors and local receptive fields, inevitably suffering from limited feature extraction capabilities and degraded classification performance. Recent Mamba-based methods, such as Mamba-HSI and HyperMamba, perform spectral–spatial feature extraction by constructing parallel and serial combinations of spatial Mamba and spectral Mamba operators, respectively. While these methods exhibit more competitive performance than Transformer-based approaches, they remain inferior to the proposed method. This phenomenon can be attributed that HSI-MFormer simultaneously benefits from both Mamba and Transformer architectures, effectively leveraging the advantages of long-range and short-range modeling to achieve spectral–spatial feature extraction. In addition, as shown in Fig. 5, the comparison methods, including SVM, 2D-CNN,

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TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods			HSI
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	CASST	3DSS-Mamba	MambaHSI	HyperMamba	-MFormer
1	98.53	98.40	97.04	92.26	99.29	97.04	99.02	**99.82**	99.46	98.93
2	98.05	99.20	98.48	83.15	98.05	98.01	99.56	99.73	99.91	**100.0**
3	98.88	**100.0**	**100.0**	99.20	**100.0**	98.71	**100.0**	99.83	**100.0**	99.84
4	97.81	98.31	98.95	93.47	96.43	99.04	97.68	**99.45**	99.10	99.38
5	98.57	99.28	99.44	99.73	**100.0**	**100.0**	99.91	**100.0**	**100.0**	**100.0**
6	92.64	92.62	**100.0**	78.76	95.89	95.38	98.29	98.96	**100.0**	**100.0**
7	91.98	94.01	93.38	91.06	96.06	95.90	97.55	**99.46**	96.84	97.63
8	91.96	89.63	94.29	86.68	91.70	96.22	96.16	95.39	**98.83**	97.77
9	85.09	83.87	93.61	89.60	94.59	95.53	96.27	94.08	97.60	**99.65**
10	93.16	94.62	96.33	89.03	97.46	99.27	99.55	**100.0**	99.63	**100.0**
11	87.01	90.53	96.84	95.31	90.56	97.89	98.20	97.36	98.02	**100.0**
12	87.25	89.38	95.46	88.63	99.28	98.54	98.38	98.08	99.27	**99.64**
13	32.35	77.83	94.24	93.12	56.40	97.23	96.68	93.31	96.20	**98.10**
14	98.91	99.30	98.60	97.92	98.44	**100.0**	99.48	**100.0**	**100.0**	**100.0**
15	85.08	**100.0**	99.85	99.66	99.66	99.39	**100.0**	**100.0**	**100.0**	**100.0**
OA (%)	91.74	93.93	96.77	91.67	95.46	97.85	98.37	98.38	98.94	**99.35**
AA (%)	90.03	93.80	97.10	91.84	94.25	97.88	98.44	98.38	98.99	**99.40**
Kappa	91.06	93.44	96.51	90.99	95.09	97.68	98.24	**100.0**	98.85	99.30

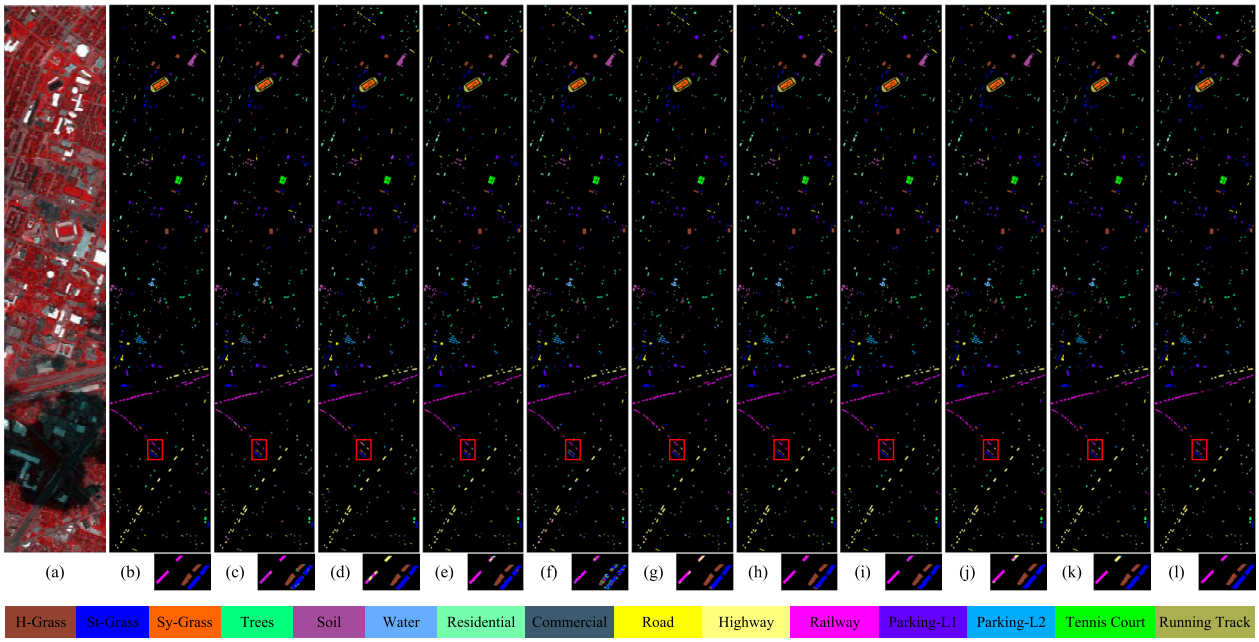


Fig. 7. Classification maps obtained by the compared methods on the Houston 2013 dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) CASST. (i) 3DSS-Mamba. (j) MambaHSI. (k) HyperMamba. (l) HSI-MFormer.

3D-CNN, and HSI-BERT, are generally prone to pronounced noise interference. In contrast, the proposed HSI-MFormer demonstrates well-maintained category boundaries with minimal mislabeling, further highlighting the superiority of the proposed method.

2) Pavia University Dataset: The classification experiments on the Pavia University dataset are executed with 5% of the reference samples. The quantitative results and visualization maps against various competitive approaches are presented in Table IV and Fig. 6. By integrating the advantages of both Mamba and Transformer structures, the proposed HSI-MFormer consistently outperforms other competitive approaches, demonstrating the highest quantities across all three metrics. Transformer-based architectures such as CASST are restricted by the single attention mechanism, which leads to a substantial accuracy reduction of 1.04% in OA, 1.66% in AA, and 1.38% in Kappa compared to the hybrid HSI-MFormer. In addition, the Mamba-based framework, which relies on a single Selective SSM, also demonstrates limited modeling capability for complex spectral–spatial patterns. Compared to the suboptimal HyperMamba, the proposed method achieves quantitative improvements of 0.11%, 0.13%, and 0.13% in OA, AA, and Kappa, respectively. According to the visualization maps in Fig. 6, both conventional and CNN-based methods present noticeable misclassification, which are limited by hand-crafted descriptors and local

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TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-LONGKOU DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods			Mamba-based Methods			HSI
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	CASST	3DSS-Mamba	MambaHSI	HyperMamba	-MFormer
1	99.17	99.92	99.33	99.94	99.80	99.46	99.85	99.63	**99.97**	**99.97**
2	81.44	94.18	93.32	93.40	91.35	99.42	99.64	99.17	99.39	**99.69**
3	55.56	86.47	92.41	94.00	89.20	98.80	96.70	91.45	**99.83**	96.86
4	97.34	98.68	98.60	99.58	99.07	99.39	98.95	99.39	99.74	**99.80**
5	55.66	76.02	91.61	86.10	87.34	99.03	96.37	**99.33**	98.73	99.19
6	98.95	99.44	98.35	99.84	98.73	98.57	98.88	**99.93**	99.81	99.76
7	99.94	99.96	99.74	**99.99**	99.78	99.98	99.94	99.82	99.95	99.95
8	85.32	90.87	90.81	90.98	93.25	94.97	96.91	94.74	**97.17**	96.04
9	79.35	83.87	89.90	92.77	89.16	86.90	93.90	**96.07**	90.03	95.90
OA (%)	95.60	97.88	98.14	98.71	98.25	99.06	99.16	99.23	99.49	**99.59**
AA (%)	83.64	92.16	94.90	95.18	94.19	97.39	97.90	97.72	98.30	**98.58**
Kappa	94.19	97.21	97.56	98.30	97.70	98.76	98.90	99.16	99.33	**99.46**

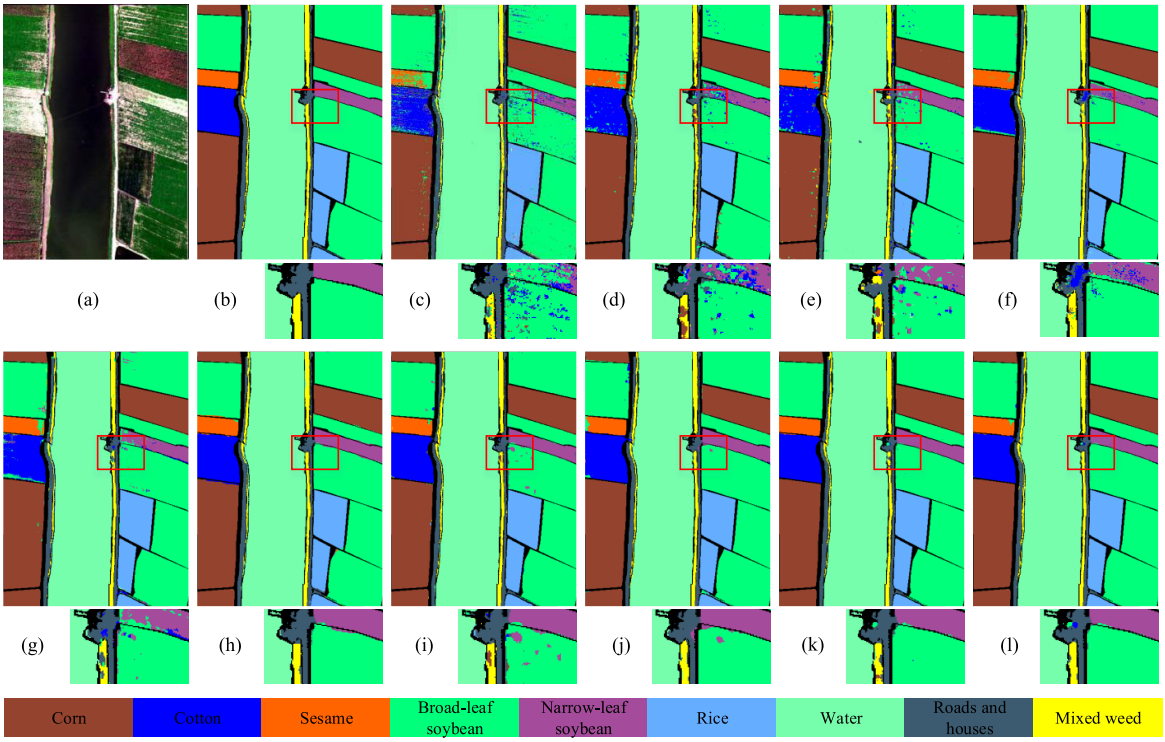


Fig. 8. Classification maps obtained by the compared methods on the WHU-Hi-LongKou dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) CASST. (i) 3DSS-Mamba. (j) MambaHSI. (k) HyperMamba. (l) HSI-MFormer.

respective kernels, respectively. Comparatively, the proposed method achieves the most consistent classification results with the ground truth, particularly excelling in categories such as Trees, Bitumen, and Self-blocking bricks. These phenomena further reveal the substantial potential of the proposed hybrid architecture for HSI classification tasks.

3) Houston 2013 Dataset: The experiments on the Houston 2013 dataset are conducted with 10% of the reference samples. The classification results and classification maps are illustrated in Table V and Fig. 7. As evidenced by the results, the proposed HSI-MFormer achieves OA, AA, and Kappa of 99.35%, 90.40%, and 90.30%. Despite the increased complexity and discrete characteristics for this testing scenario, HSI-MFormer performs long-short spatial–spectral contextual modeling based on the selective SSMs and attention mechanism, demonstrating significant performance advantages. Limited by its modeling capacity, the classical HSI-BERT inevitably yields degraded classification results, with reductions of 7.68% in OA, 7.56% in AA, and 8.31% in Kappa compared to HSI-MFormer. As shown in the visualization maps in Fig. 7, the proposed HSI-MFormer further provides the most precise prediction details, exhibiting superior object integrity. These experimental results indicate that the proposed method effectively leverages the complementary strengths of Transformer and Mamba, significantly enhancing spectral–spatial feature representation.

4) WHU-Hi-LongKou Dataset: Table VI and Fig. 8 present the corresponding classification results and visualization maps. The experiments on the WHU-Hi-LongKou dataset are conducted with 1% of the reference samples. As demonstrated

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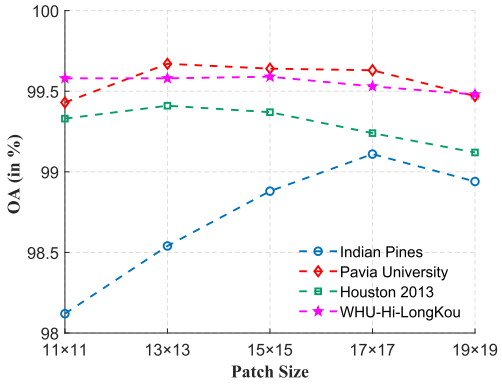


Fig. 9. Sensitivity analysis for the proposed method with different sizes of input patches.

by the results, the proposed HSI-MFormer achieves the highest OA, AA, and Kappa of 99.59%, 90.58%, and 99.46%, respectively. Due to the inherent limitations in feature extraction, conventional methods, and CNN-based approaches exhibit relatively weaker performance. By integrating the complementary strengths of selective SSMs and attention mechanism, HSI-MFormer outperforms both Mamba-based and Transformer-based methods. Compared to the suboptimal HyperMamba, the proposed HSI-MFormer achieves quantitative improvements of 0.10%, 0.28%, and 0.13% in OA, AA, and Kappa, respectively. As illustrated in the visualization maps in Fig. 8, HSI-MFormer further provides the most accurate prediction details, which aligns with the quantitative results in Table VI. These observations underscore that the proposed method maintains its effectiveness and superiority in high-resolution scenarios.

E. Discussion and Analysis

1) Sensitivity to Different Input Patch Sizes: Ablation experiments are performed to discuss the impact of different input patch sizes on classification performance, as illustrated in Fig. 9. The input patch ranges from 11 × 11 to 19 × 19 with a growth interval of 2. As the patch size increases, all datasets display a similar trend, with performance either consistently improving or relatively stabilizing. However, the OA gradually decreases upon reaching the identified point (15 × 15 for Houston 2013, Pavia University, and WHU-Hi-LongKou datasets, 17 × 17 for the Indian Pines dataset). This decline may be attributed to the introduction of redundancy or noise with excessively large patches, leading to information aliasing. Although the patch size of 17 × 17 proves optimal for the Indian Pines dataset, the 15 × 15 still maintains near-optimal performance. Based on these observations, a patch size of 15 × 15 is suitable for all datasets in our experiments.

2) Sensitivity to Different Embedding Dims and Depths: Ablation experiments are conducted to investigate the sensitivity of the proposed HSI-MFormer to stage depth N and feature embedding dimensions. Figs. 10 and 11 illustrate the performance variations on three datasets in terms of OA (%), AA (%), and Kappa (%). The range of depth is set to $\{1, 2, 3, 4, 5\}$, and the embedding dim covers an interval of $\{8, 16, 32, 48, 64\}$. As can be observed, lower depths and embedding dimensions are sufficient to achieve promising classification performance. Conversely, excessive dimensions and deeper depths provide limited accuracy improvements. This limitation may be attributed to the insufficient sample size of the hyperspectral data, which fails to support the model's learning capacity and potentially leads to overfitting. Accordingly, the proposed HSI-MFormer is constructed as a lightweight structure, with the embedding dimension set to 32 and the stage depth determined as 3.

3) Sensitivity to Different Kernel Scales: The kernel scales in MTG correspond to different receptive fields, allowing the model to capture spectral–spatial patterns at varying granularities, which is crucial for extracting multiscale information from hyperspectral data. As illustrated in Fig. 12, the [5, 7, 9] configuration consistently achieves relatively favorable results in terms of OA, AA, and Kappa coefficient. The scale of 5 is adept at capturing fine-grained spectral–spatial details, while the scale of 9 excels at extracting broader semantic and structural variations. The intermediate scale of 7 serves a balancing role, enhancing feature complementarity and robustness. This configuration maintains sufficient scale diversity, ensuring that each kernel contributes distinct and complementary information to the feature representation.

To explore the possibility of better parameter combinations, we further investigate the sensitivity of MTG to different scale configurations. As depicted in Fig. 12, the configuration [3, 4, 5], characterized by a scale difference of 1 between adjacent scales, consistently underperforms across all datasets. The possible reason might be that excessively similar kernel scales may lead the model to capture redundant information, reducing the diversity and complementarity of features extracted at different scales. In contrast, the configuration [3, 7, 11] with a scale difference of 4 achieves superior performance on the Houston 2013 and WHU-Hi-LongKou datasets. This improvement may be attributed to the enhanced feature discriminability enabled by sufficient scale variation. These findings suggest that configurations with larger scale differences are relatively advantageous for complex and high-resolution HSI scenarios, which can be prioritized in our further works.

4) Effectiveness of ITE and CME: This section systematically investigates the effectiveness of the ITE and CME experts. The classification performance on three datasets is listed in Table VII. It can be observed that both the individual ITE and CME independently achieve substantial classification performance, revealing the efficacy of the intrinsic self-attention mechanism and SSMs. Moreover, the integration of these two experts further enhances the classification capability, demonstrating strong complementarity and robustness across all datasets. This improvement can be attributed to that the hybrid architecture can integrate the strengths of the Transformer and Mamba, thus effectively capturing multiscale spectral–spatial dependencies in short and long ranges. Generally, both ITE and CME contribute to the classification and all these components corporately yield the most superior results in different scenarios.

5) Effectiveness of MTG: This section systematically evaluates the classification performance with and without the MTG

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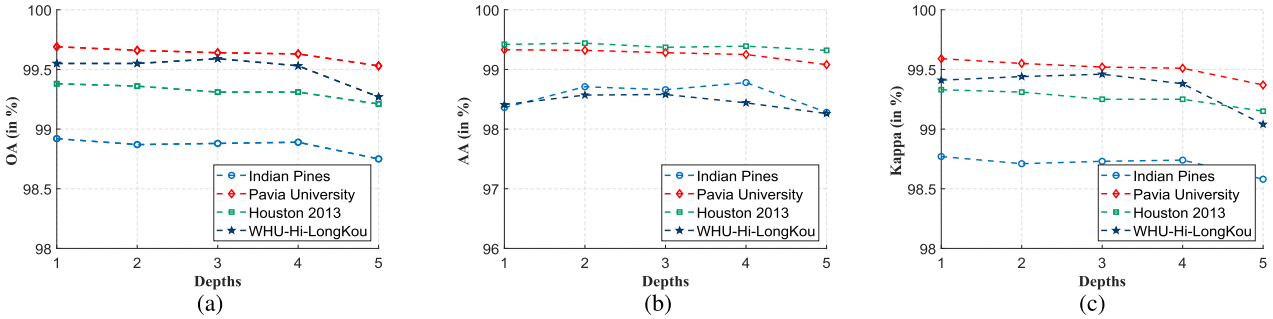


Fig. 10. Sensitivity analysis for the proposed method with different stage depths in terms of (a) OA, (b) AA, and (c) Kappa.

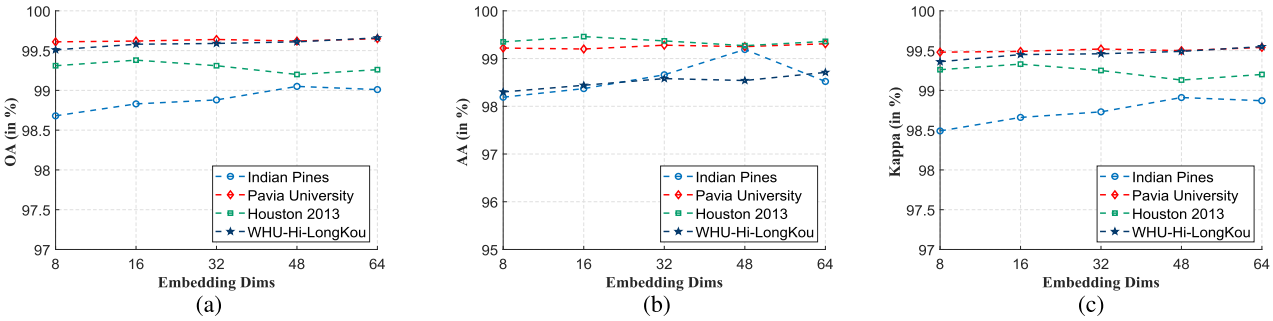


Fig. 11. Sensitivity analysis for the proposed method with different feature embedding dimensions in terms of (a) OA, (b) AA, and (c) Kappa.

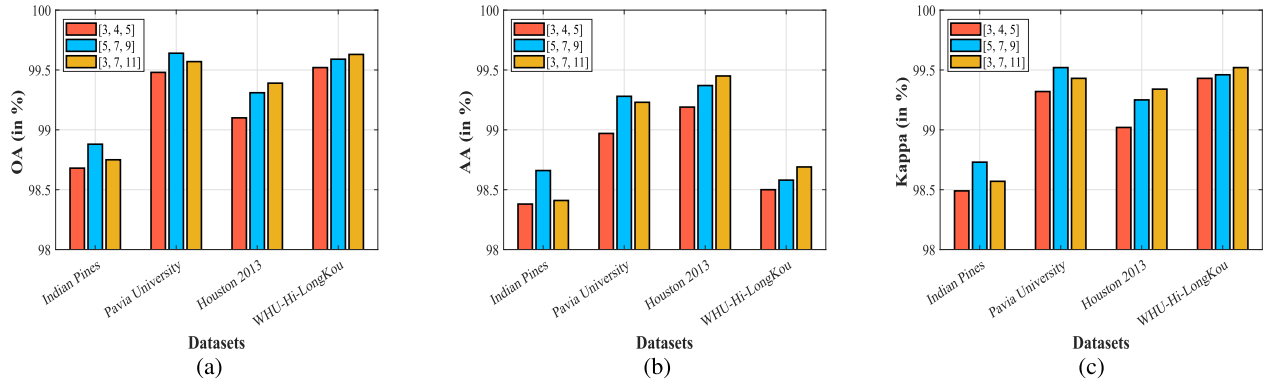


Fig. 12. Sensitivity analysis for the proposed method with different scales in terms of (a) OA, (b) AA, and (c) Kappa.

module on four HSI datasets. The results in terms of OA, AA, and Kappa are illustrated in Table VIII. The "w/o MTG" refers to setting the scales of s, m, and l to be identical, leading the model to operate within a single scale. Compared to the w/o MTG configuration, the classification performance of HSI-MFormer with MTG demonstrates noticeable improvements, with average increases of 0.29% in OA, 0.48% in AA, and 0.34% in Kappa. By constructing multiple spatial–spectral token groups across diverse scales, MTG facilitates the model to comprehensively capture spectral-spatial details and structures at varying granularities, which is essential to address the high variability and complexity inherent in hyperspectral data. These results highlight the pivotal role of the MTG module in multiscale spatial–spectral information extraction, significantly enhancing the model's performance in HSI classification tasks.

F. Sensitivity to Different Training Ratios

To evaluate the robustness of the proposed HSI-MFormer framework, extensive experiments are conducted by varying the proportions of training samples on the Indian Pines, Pavia University, Houston 2013, and WHU-Hi-LongKou datasets. Fig. 13 presents the performance variations, with the proposed HSI-MFormer prominently highlighted by red curves. The percentages of the selected training samples on four datasets cover an interval of $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, $\{0.5\%, 1.0\%, \ldots, 5.0\%\}$, $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, and $\{0.1\%, 0.2\%, \ldots, 1.0\%\}$, respectively. The results reveal a consistent improvement in classification accuracy for HSI-MFormer as the proportion of training samples increases. Meanwhile, HSI-MFormer consistently outperforms other competitive methods across most percentages on the four testing datasets, further highlighting the robustness and effectiveness of the proposed hybrid architecture.

G. Analysis of Computational Complexity

As illustrated in Table IX, the computational complexity of the proposed HSI-MFormer is investigated, focusing on model parameters, Flops, training time, and testing time. Parameters

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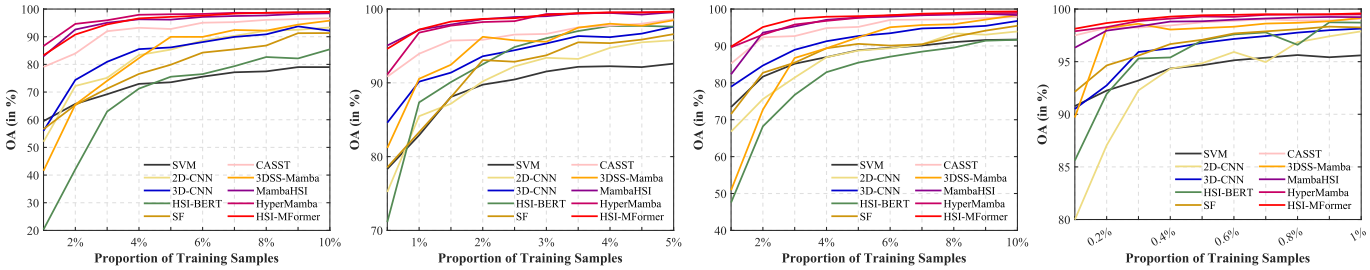


Fig. 13. Classification performance of the compared methods with different proportions of labeled data on four datasets. (a) Indian Pines. (b) Pavia University. (c) Houston 2013. (d) WHU-Hi-LongKou.

TABLE VII

EFFECTIVENESS ANALYSIS OF ITE AND CME ON ALL DATASETS IN TERMS OF OA (%), AA (%), AND KAPPA (%)

ITE	CME	Indian Pines			Pavia University			Houston2013			WHU-Hi-LongKou		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
✓		98.75	98.23	98.57	99.52	98.99	99.36	99.07	99.18	99.00	99.35	98.25	99.32
	✓	98.3	97.64	98.06	99.55	98.98	99.4	99.22	99.3	99.16	99.44	98.37	99.43
✓	✓	**99.01**	**98.91**	**98.87**	**99.67**	**99.38**	**99.57**	**99.35**	**99.40**	**99.30**	**99.59**	**98.58**	**99.46**

TABLE VIII

EFFECTIVENESS ANALYSIS OF MTG ON ALL DATASETS IN TERMS OF OA (%), AA (%), AND KAPPA (%)

	Indian Pines			Pavia University			Houston2013			WHU-Hi-LongKou		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
w/o MTG	98.54	98.18	98.33	99.38	98.67	99.18	99.05	99.16	98.98	99.48	98.33	99.32
HSI-MFormer	**99.01**	**98.91**	**98.87**	**99.67**	**99.38**	**99.57**	**99.35**	**99.40**	**99.30**	**99.59**	**98.58**	**99.46**

TABLE IX

COMPUTATIONAL COMPLEXITY ANALYSIS OF VARIOUS COMPARISON APPROACHES IN TERMS OF PARAMETERS, FLOPS, TRAINING TIME, AND TESTING TIME

Metrics	CNN-based Methods		Transformer-based Methods			Mamba-based Methods			HSI-MFormer
	2D-CNN	3D-CNN	HSI-BERT	SF	CASST	3DSS-Mamba	MambaHSI	HyperMamba	
Params (M)	0.0132	0.0123	0.3773	0.1133	6.5316	0.0105	0.0370	0.0861	0.1322
Flops (G)	0.0002	0.0017	6.8135	0.0163	0.9893	0.0230	13.030	0.0031	0.0689
Training time (s)	4.47	4.02	62.13	97.28	45.82	168.85	233.61	20.12	81.26
Testing time (s)	0.34	0.44	20.18	2.4	2.62	3.95	0.02	1.54	3.89

refer to the number of learnable parameters in the entire network. Flops denote the number of floating-point operations executed by the model. Although CNN-based methods exhibit relatively fewer parameters, lower FLOPs, and training and testing times, their ability to capture complex spatial–spectral information is limited by local receptive fields, leading to suboptimal classification accuracy. Transformer-based architectures suffer from quadratic computational complexity due to the inherent MHSA mechanism, resulting in higher training and testing times (e.g., 20.18 s testing time for HSI-BERT). Compared to the dual-branch Transformer-based CASST, the proposed HSI-MFormer significantly reduces computational parameters and Flops. Mamba-based methods (e.g., MambaHSI, HyperMamba) demonstrate lower testing times owing to the linear computational complexity of SSMs. The higher training time (e.g., 233.61 s for MambaHSI) primarily arises from large epoch settings and direct processing of global HSI data rather than patch-wise inputs. Comparatively, the proposed HSI-MFormer integrates the complementary strengths of both Transformer and Mamba experts, its computational complexity is slightly higher than that of purely Mamba-based methods such as HyperMamba. The significant improvement in classification accuracy across all datasets (see Tables III–VI) indicates that the additional computational cost is reasonable and acceptable. In summary, the proposed HSI-MFormer achieves an excellent tradeoff between computational efficiency and classification accuracy, highlighting substantial potentiality and superiority in HSI classification tasks.

V. CONCLUSION

In this study, we propose a novel hybrid Mamba-Transformer experts architecture HSI-MFormer, which leverages both SSMs and self-attention mechanisms to capture multiscale spatial–spectral information in both short-range and long-range dependencies. The Transformer-based ITE performs short-range dependency modeling within token groups at each scale, which effectively captures the fine-grained spatial–spectral patterns while minimizing computational complexity. The Mamba-based CME expert focuses on long-range sequence modeling, which significantly captures the complementarity between token groups across different scales with linear computational complexity. Extensive experiments

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indicate that the proposed HSI-MFormer efficiently breaks the performance and efficiency limitations in state-of-the-art Transformer-based and Mamba-based HSI architectures. Future research will focus on developing adaptive strategies to optimize the integration of different network architectures and extend their application to more challenging scenarios.

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